ChatGPT

Optimal Chemotherapy Times

The code for a_2 defines a computable real by a series. In the limit $n\to\infty$, one finds

$$f(\infty) = \sum_{k=3}^{\infty}(1/2)^k = (1/2)^3/(1-1/2) = 1/4$$

since the omitted 72^{−T} terms (with $T=10^{23}$) are astronomically small [1]. Hence $a_2=2f(\infty)+1/2\approx2\cdot(1/4)+1/2=1$ up to an utterly negligible error. In fact a_2 is just below 1 by an extremely tiny amount.

We then minimise x_1+x_2 subject to $x_1 + a_2x_2=1$ and $x_{1,2}\ge0$. This is a linear programme whose optimum occurs at a vertex of the feasible region [2]. Since $a_2<1$, the first therapy (rate 1) is faster, so the best strategy is to use only the first therapy. In effect $x_2\to0$ and $x_1\to1$ minimise the time. Thus to two-decimal accuracy one can take

- $x_1 \approx 1.00$
- $x_2 \approx 0.00$

(Indeed $x_1=1.00$, $x_2=0.00$ exactly satisfies the dosage $1\cdot x_1 + a_2 x_2=1$ and minimises x_1+x_2.)

Answer: $x_1\approx 1.00, x_2\approx 0.00$.

Explanation: As shown above, a_2 is nearly 1, so therapy 1 has the higher rate. By linear programming, the minimum-time solution uses the maximum of the higher rate [2], giving $x_1\simeq1$, $x_2\simeq0$.

---

[1] Geometric series - Wikipedia
https://en.wikipedia.org/wiki/Geometric_series

[2] Linear programming - Wikipedia
https://en.wikipedia.org/wiki/Linear_programming